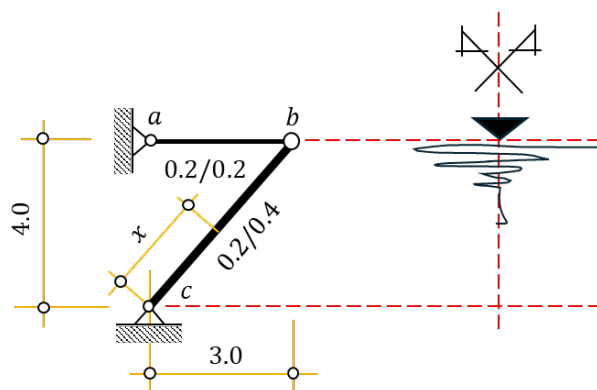


1: За систем на скици услед дејства течности  $\gamma = 10 \text{ kN/m}^3$  срачунати и нацртати:

- статичку и кинематичку класификацију,
- реакције у ослоњцима и силе у пресецима,
- на делу "с-б", положај попречног пресека са максималном вредности  $M$  савијања.

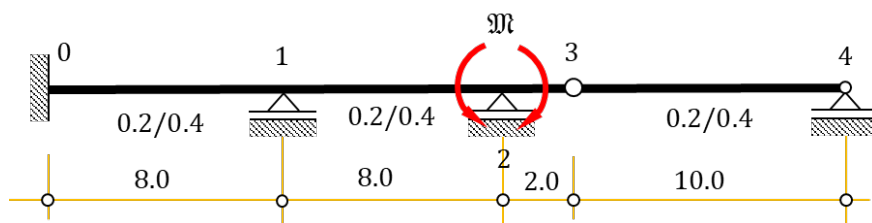
Подаци:  $E = 3 \cdot 10^7 \frac{\text{kN}}{\text{m}^2}$ ,



2: За носач приказан на скици, срачунати и нацртати:

- вредност пара момената  $M$  који изазива релативно обртање попречног пресека лево и десно у зглобу "3" од  $\Delta\varphi_3 = 0.01 \text{ rad}$ ,
- дијаграм момената савијања,
- вертикално померање чвора "3".

Подаци:  $E = 3 \cdot 10^7 \frac{\text{kN}}{\text{m}^2}$ , занемарити  $N$  и  $T$  - силе на деформацију.



Предметни наставник:

проф. др Илија М. МИЛИЧИЋ, дипл.инж.грађ.

17.04.2026

$$\begin{aligned} Z_0 + Z_L + Z_R + Z_S &\stackrel{>}{\stackrel{<}{=}} 2R \\ \downarrow \\ \frac{1}{4} + 0 + 0 + 2 &= 2.3 \end{aligned}$$

$G = G$   $\Rightarrow$  СТАТИЧКИ ОДРЕДЕН  
КИНЕМАТИЧКИ  
ПРОСТО СТАБИЛЕН

## \* КОРЕСПОЛЕНТИХ ПЛОМА

1° ВАРУЖАТА

№ 44444

$$Z_s + Z_k \geq 2k - 3$$

$$5 + 2 = 2.5 - 3$$

$$7 = 7$$

OK

20 НАЧМН

$$2 \cdot z_7 \begin{matrix} \geq \\ \equiv \\ \leq \end{matrix} 3z_p - 3$$

$$2.3 = 3.3 - 3$$

$$6 = 6$$

OK

2º BAPUNATA

10H4CH

$$Z_S + Z_K \stackrel{>}{<} 2k - 3$$

$$6 + 3 = 2.6 - 3$$

$$g = g$$

OK

2p ~~1~~ 4M 11

$$2.72 \approx 37p-3$$

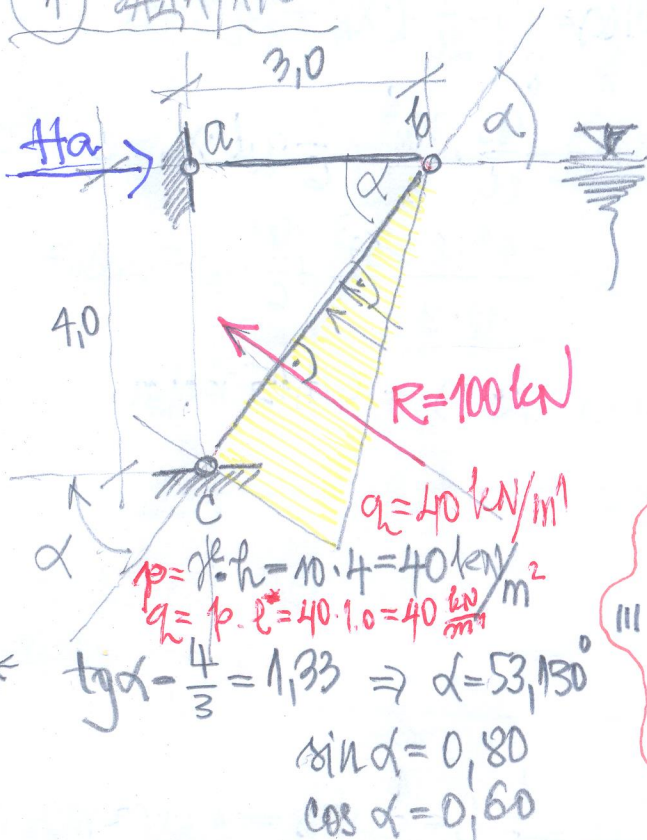
$$2.3 = 3.3 - 3$$

$$6 = 6$$

OK

\* ВНУТРИШНЕ КИНЕМАТИЧКИ ПРОСТО СТАБИЛАН ПОСЛУЖИТАТО.

## 1) ЗАДАЧА

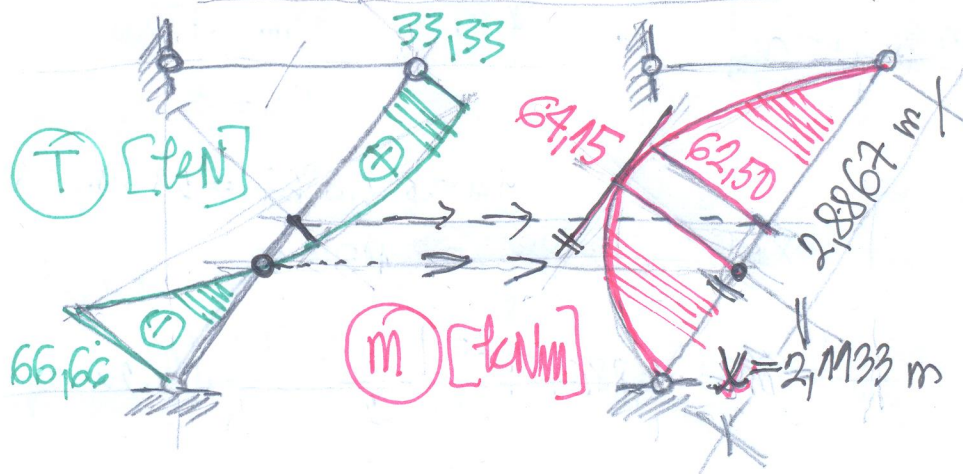
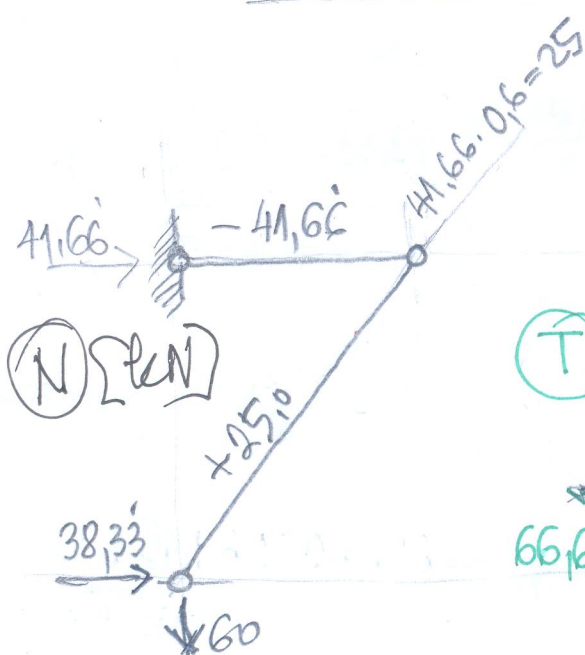


$$\Sigma H = 0: H_a + H_c - 100 \cdot 0,8 = 0$$

$$\Sigma V = 0: V_c + V_a + 100 \cdot 0,6 = 0$$

$$\sum M_c = 0: -H_a \cdot 4,0 + 100 \cdot 0,8 \cdot 0,8 \cdot 1,66 + 100 \cdot 0,6 \cdot 0,6 \cdot 1,66 = 0$$

Решаю:  $H_a = 41,66 \text{ кН}$

$$H_0 = 38,33 \text{ kN}$$
$$Y_c = -60 \text{ kN}$$




OK! ☺



Thompson

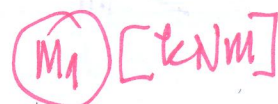


$$Z_0 + Z_u + Z_k + Z_s \geq 2k$$

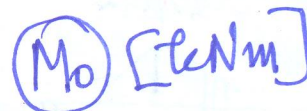
$$5 + 1 + 2 + 4 > 2.5$$

$$m = \sum z - 2k = 12 - 10 = 2 \times \text{CHC.}$$

$X_1 = 1.0$  ←



A hand-drawn diagram on a white background. It features a horizontal line and a vertical line intersecting at a point. The horizontal line is drawn with a blue pen, and the vertical line is drawn with a red pen. The intersection point is marked with a small red dot. The lines extend slightly beyond the intersection point.


$$S_{11} = \frac{1}{3} \cdot 1 \cdot 1 \cdot 8 = \frac{8}{3}$$

$$S_{12} = \frac{1}{6} \cdot 1 \cdot 1 \cdot 8 = \frac{8}{6}$$

$$\sigma_{22} = 2 \cdot \frac{1}{3} \cdot 1 \cdot 1 \cdot 8 = \frac{16}{3}$$

$$\mathcal{F}_1 = \emptyset$$

$$\delta L_0 = \frac{-1}{6} \cdot 1 \cdot m^4 \cdot 8 = -\frac{8}{6} m^4$$

$$\textcircled{10} \quad \frac{8}{3}x_1 + \frac{8}{6}x_2 + 0 = 0$$

$$(2^o) \quad \frac{8}{6}x_1 + \frac{16}{3}x_2 - \frac{8}{6}m = 0$$

$$\Rightarrow \textcircled{10} \quad x_1 = -\frac{\frac{8}{6}x_2}{\frac{8}{8}} = -\frac{24}{48}x_2 \quad \boxed{x_1 = -\frac{1}{2}x_2}$$

$$\textcircled{20} \quad -\frac{8}{12}x_2 + \frac{16}{3}x_2 = \frac{8}{6}m^4$$

$$\frac{56}{12} x_2 = \frac{8}{6} \text{ m}^4 \quad x_2 = \frac{\frac{8}{6}}{\frac{56}{12}}$$

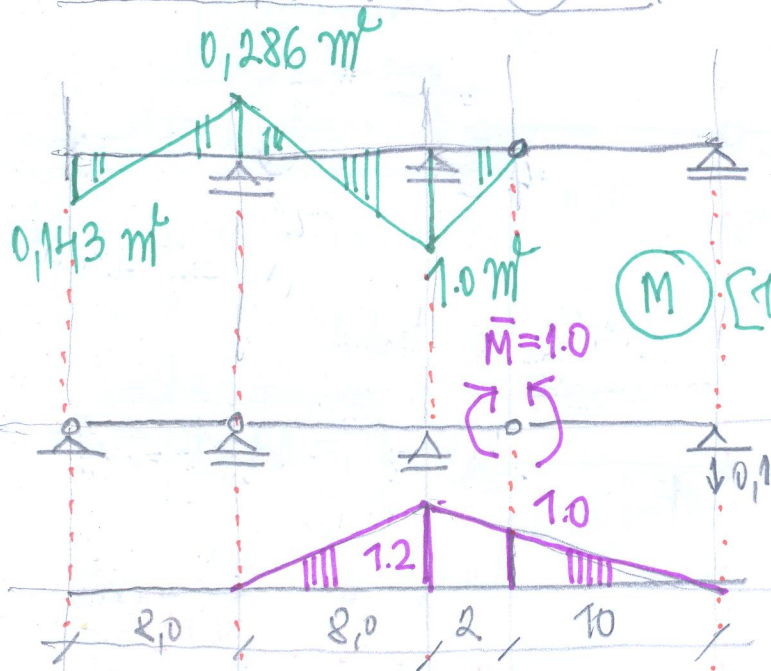
$$x_2 = \frac{96 \text{ m}^L}{326} = \frac{1}{7} \text{ m}^L = 0,286 \text{ m}^L$$

$$x_1 = -\frac{1}{2} \cdot \frac{2}{7} \text{ м} = -\frac{1}{7} \text{ м} \quad * \text{ ПЕРВАЯ}$$

$$x_1 = -0,143 \text{ m} ; \quad x_2 = 0,286 \text{ m}$$

КОШАЧНИ АМПЛЕРАМ

(M) y φ-ш (M)



$$M = M_0 + M_1 \cdot X_1 + M_2 \cdot X_2$$

$$M = 0 + 1 \cdot (-0.143 \text{ m}^2) + 0.286 \text{ m}^2$$

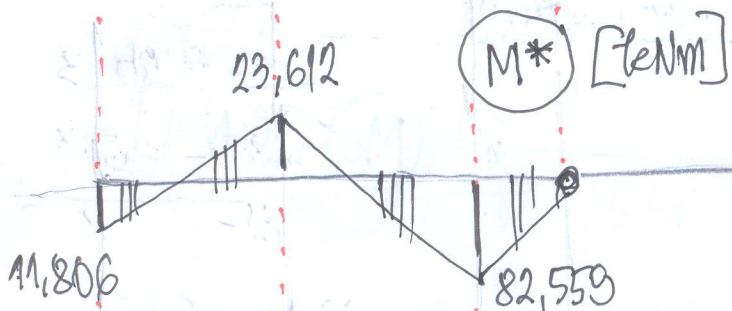
$$M = -0.143 \text{ m}^2$$

$$EI = 3 \cdot 10^7 \cdot \frac{0.2 \cdot 0.4^3}{12} = 32000$$

(M-bar) [kNm]

$$EI \Delta \varphi_3 = \int M \cdot \bar{M} \Delta \varphi_3 ds = \frac{1}{6} \cdot 1.2 (-2 \text{ m}^2 + 0.286 \text{ m}^2) \cdot 8.0 - \frac{1}{6} \text{ m}^2 \cdot (2 \cdot 1.2 + 1) \cdot 2.0$$

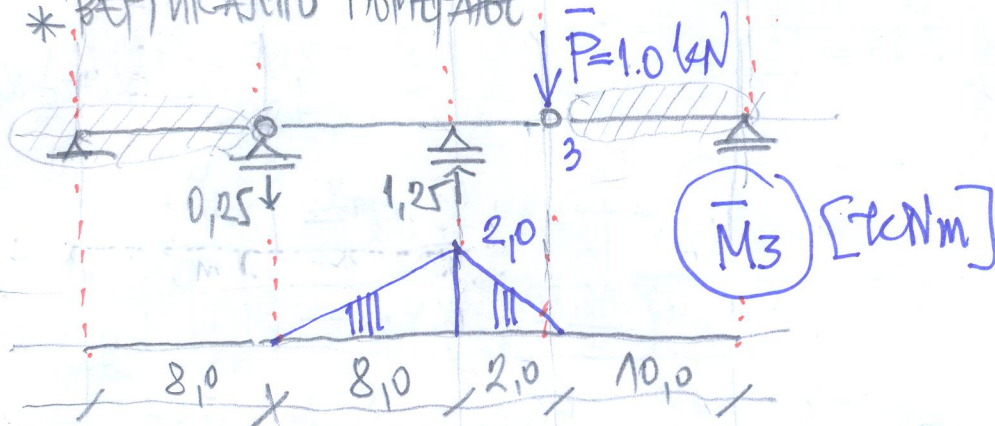
$$EI \Delta \varphi_3 = -3.876 \text{ m}^2 \Rightarrow \Delta \varphi_3 = -\frac{3.876 \text{ m}^2}{EI} = -\frac{3.876 \text{ m}^2}{32000} = 0.01 \text{ rad}$$



$$M^* = 0.01 \cdot 32000 \cdot \frac{1}{3.876}$$

$$M^* = 82.559 \text{ kNm}$$

\* ВЕРТИКАЛНО ПОМЕРАЊЕ



$$EI v_3 = \int M^* \cdot \bar{M}_3 ds = \frac{1}{6} 2 (2 \cdot -82.559 + 23.612) \cdot 8.0 - \frac{1}{3} 2 \cdot 82.559 \cdot 2.0 = -487.428$$

$$v_3 = -\frac{487.428}{32000} \cdot 1000 = -15.232 \text{ mm}$$

РЕУЊЕ

$$v_3 = -15.232 \text{ mm}$$